

Effects of consultations and interventions in a pharmacist-led outpatient clinic on duration of treatment and adverse events with osimertinib

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Abstract

Purpose: Osimertinib, which is a key treatment for patients with epidermal growth factor receptor gene mutation-positive non-small cell lung cancer (EGFR mt NSCLC), causes intractable adverse events for some patients. The objective of this study was to assess the impact of pharmacist consultation in a pharmacist-led outpatient clinic (PLOC) and the effectiveness of pharmacist interventions on osimertinib treatment.

Patients and Methods: This observational cohort study included patients who started osimertinib for EGFR mt NSCLC at Aichi Medical University Hospital between April 2018 and December 2021. The duration of treatment and occurrence of adverse events were compared according to whether they consulted a PLOC pharmacist, and whether they received pharmacist intervention. This study was approved by the ethical review board of the university (approval no. 2019-203).

Results: The median duration of treatment was significantly longer for the patients who consulted with the PLOC pharmacist than for those who did not (561 vs 203 days, hazard ratio 0.40, $p < 0.001$). The median duration of treatment was significantly longer for patients who received pharmacist intervention than for those who did not (774 vs 237 days, hazard ratio 0.39, $p < 0.001$). The discontinuation rate was significantly lower in patients who consulted a PLOC pharmacist than for those who did not (73% vs 97%, $p = 0.008$). However, the rates and reason for osimertinib discontinuation or dose reduction did not differ between groups.

Conclusion: PLOC consultation and intervention for the treatment of adverse events might have led to extending the duration of osimertinib treatment.

Keywords

Pharmacist-led outpatient clinic, tyrosine kinase inhibitor, osimertinib, non-small cell lung cancer, duration of treatment

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Introduction

Osimertinib, a third-generation oral epidermal growth factor receptor tyrosine kinase inhibitor (EGFR-TKI), selectively and irreversibly suppresses active mutated EGFR and resistant mutated EGFR T790M.¹ EGFR gene mutations are the major driver gene mutations in 45% of Japanese lung adenocarcinomas,² which is the most frequent histological type of lung cancer in Japan. Osimertinib, which was first approved by the Food and Drug Administration in 2015, followed by the European

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Medicines Agency in 2016. It is now a key treatment for patients with EGFR gene mutation-positive non-small cell lung cancer (NSCLC). In Japan, the Ministry of Health and Welfare approved it in March 2016 for T790M mutation-positive advanced NSCLC that has progressed during or after treatment with a first- or second-generation EGFR-TKI. Additional indications for the first-line treatment of EGFR mutation-positive advanced NSCLC were approved in August 2018 and for postoperative adjuvant therapy in August 2022.

In a global phase III clinical trial for patients with advanced NSCLC (FLAURA trial), osimertinib significantly prolonged progression-free survival compared with the first-generation EGFR-TKIs and also showed efficacy for central nervous system metastases.³ The final analysis of the FLAURA trial also showed that osimertinib significantly prolonged overall survival compared with other EGFR-TKIs.⁴ Although osimertinib has fewer grade 3 or higher adverse events than first-generation EGFR-TKIs,³ some patients still experience significant adverse events, including diarrhea, paronychia, oral mucositis, and acne-like skin rashes, leading to treatment withdrawal or dose reduction.

To date, several reports have shown that pharmacist intervention can have positive effects on patients treated with other anti-cancer agents in an outpatient setting.^{5–7} They showed that pharmaceutical outpatient services for anticancer drugs contribute to a reduction in severity,^{6,8} decrease the frequency,^{9,10} and assist in identifying adverse events.¹¹ Aichi Medical University Hospital started an outpatient pharmacist service for lung cancer patients treated with oral anticancer drugs in 2017. Patients who received osimertinib treatment were added to the target population of the pharmacist-led outpatient clinic (PLOC) in 2019.

In the present study, the efficacy of PLOC consultation and pharmacist interventions on the duration of treatment and the occurrence of adverse events in patients who received osimertinib were evaluated.

Methods

Patients

Consecutive patients who started treatment with osimertinib for NSCLC at the hospital between 1 April 2018 and 31 December 2021 and received continuous follow-up of the treatment in an outpatient setting were included.

Intervention/service

In the PLOC, patients consulted with a pharmacist before a doctor's appointment. They were sometimes examined by a doctor in the presence of the pharmacist, who judged it necessary. Patients were provided with treatment

recommendations based on pharmacist assessments. When necessary, the pharmacist and physician collaborated to determine the most appropriate treatment. Patients also received instructions on how to deal with adverse events, such as skin care and the use of medicines for management of adverse events. All patients who started osimertinib consulted in the PLOC after 1 August 2019, when osimertinib was added to the target drug list of the PLOC. For patients who had started osimertinib before that date, the PLOC consultation was available at the request of doctors or patients.

Data collection

We collected patient demographic data, Eastern Cooperative Oncology Group (ECOG) performance status, prior EGFR-TKI use, EGFR mutation type, and occurrence of adverse events (diarrhea, oral mucositis, dry skin, paronychia, and skin rash). Additionally, details of pharmacist interventions were recorded. The observation period was set until 31 December 2022. Adverse events were assessed using the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0.

Outcomes

The primary outcome was the duration of treatment with osimertinib, and the secondary outcome was the frequency of occurrence of adverse events. These outcomes were compared according to whether patients consulted the PLOC pharmacist and whether they received pharmacist intervention. Patients were considered to have 'consulted the PLOC' if they had been interviewed at least once in the PLOC. Furthermore, they were considered to have 'received pharmacist intervention' if they had received it at least once.

Statistical analysis

Statistical analyses were performed using the statistical software EZR version 1.63,¹² with the Mann-Whitney U test for comparisons of quantitative data and the Fisher's exact probability test for qualitative data. The duration of treatment with osimertinib was assessed using the Kaplan-Meier method, and the log-rank test and the cox regression hazard model were used for comparisons between groups. The significance level was set at a p-value of less than 0.05 for all analyses.

Ethical considerations

This study was conducted in compliance with the Declaration of Helsinki 1975, revised Brazil 2013, and the Ethical Guidelines for Medical and Biological Research Involving Human Subjects in Japan. This study was approved by the ethical review board of the Aichi

Medical University School of Medicine (approval number: 2019-203), and opt-out consent was offered with the permission of the board because no intervention beyond routine clinical practice was included.

Results

Patients' background characteristics

Ninety-nine patients started treatment with osimertinib during the study period, of whom 13 were excluded from the analysis, and 86 were analyzed. Reasons for exclusion from the analysis were 8 patients without outpatient visits for their own reasons, 2 patients with irregular visits and poor compliance, 2 patients transferred to another hospital during osimertinib treatment, and 1 patient who received osimertinib again. Of the 86 patients analyzed, 55 consulted a pharmacist in the PLOC, and 31 did not. Of the patients who consulted a PLOC pharmacist, 38 received pharmacist intervention. Of the patients who did not consult a PLOC pharmacist, one received intervention by the pharmacist, which was by telephone consultation.

Table 1. Patients' characteristics.

	With consultation (n = 55)	Without consultation (n = 31)	p value
Age, y			
median (range)	70 (37–84)	72 (42–84)	0.181 ^a
Sex			
female/male	39/16	20/11	0.630 ^b
Body weight, kg			
median (range)	51 (33.6–96.2)	53 (36.4–73.8)	0.910 ^a
ECOG-PS			0.142 ^b
0	22	11	
1	23	9	
2	6	4	
3	3	2	
Missing data	1	5	
Previous EGFR-TKI			
yes/no	14 / 41	15 / 16	0.036 ^b
Type of EGFR mutation			0.833 ^b
del 19	28	16	
del 19 + T790M	5	3	
L858R	17	9	
L858R + T790M	1	2	
Others	4	1	

^aMann-Whitney U test.

^bFisher's exact test.

ECOG-PS: eastern cooperative oncology group-performance status,
EGFR-TKI: epidermal growth factor receptor-tyrosine kinase inhibitor.

There were no differences in age, sex, body weight, PS, and type of EGFR mutation between the two groups of patients with and without PLOC pharmacist consultation. More patients who consulted a PLOC pharmacist had never been treated with a first- or second-generation EGFR-TKI before osimertinib treatment than those who did not ($p < 0.036$, Table 1).

Duration of treatment

The impact of PLOC pharmacist consultations on the duration of osimertinib treatment is shown in Figure 1. The median treatment duration was significantly longer for patients who consulted a PLOC pharmacist than for those who did not (561 vs 203 days, hazard ratio (HR): 0.40, $p < 0.001$, Figure 1(A)). Patients who received pharmacist intervention had a significantly longer median duration of treatment than those who did not (774 vs 237 days, HR: 0.39, $p < 0.001$, Figure 1(B)). The median duration of treatment of the patients who consulted a PLOC pharmacist and received pharmacist intervention, who consulted with a pharmacist but did not receive pharmacist intervention, and who neither consulted nor received pharmacist intervention was 774 days, 308 days and 209.5 days, respectively ($p < 0.001$, Figure 1(C)).

The proportion of patients who required osimertinib dose reduction was not significantly different between the patients who consulted a PLOC pharmacist and those who did not, 51% and 42%, respectively. The reasons for dose reduction were diarrhea, paronychia, skin rash, dry skin and oral mucositis in 12 patients, other adverse events in 12 patients, and others in 4 patients among the patients who consulted a PLOC pharmacist, compared with 4, 7 and 2 patients among those who did not, respectively.

Of the patients who consulted a PLOC pharmacist, 40 discontinued osimertinib treatment during the observation period. Reasons for discontinuation were disease progression in 24 patients, death in 1 patient, adverse events in 8 patients, and other reasons in 7 patients. Details of adverse events that led to discontinuation included osimertinib-induced pneumonia (including suspected) in 4 patients, decreased blood cell counts in 2 patients, paronychia in 1 patient, and itching and palpitations in 1 patient. In the patients who did not consult a PLOC pharmacist, osimertinib treatment was discontinued during the observation period in 30 patients. Reasons for discontinuation were disease progression in 14 patients, death in 3 patients, and adverse events in 9 patients. Details of adverse events that caused osimertinib discontinuation were drug-induced pneumonia in 6 patients, decreased blood cell counts in 1 patient, general malaise in 1 patient, and skin rash in 1 patient. There was no significant difference in the reasons for discontinuation between the patients who consulted a PLOC pharmacist and those who did not ($p = 0.400$).

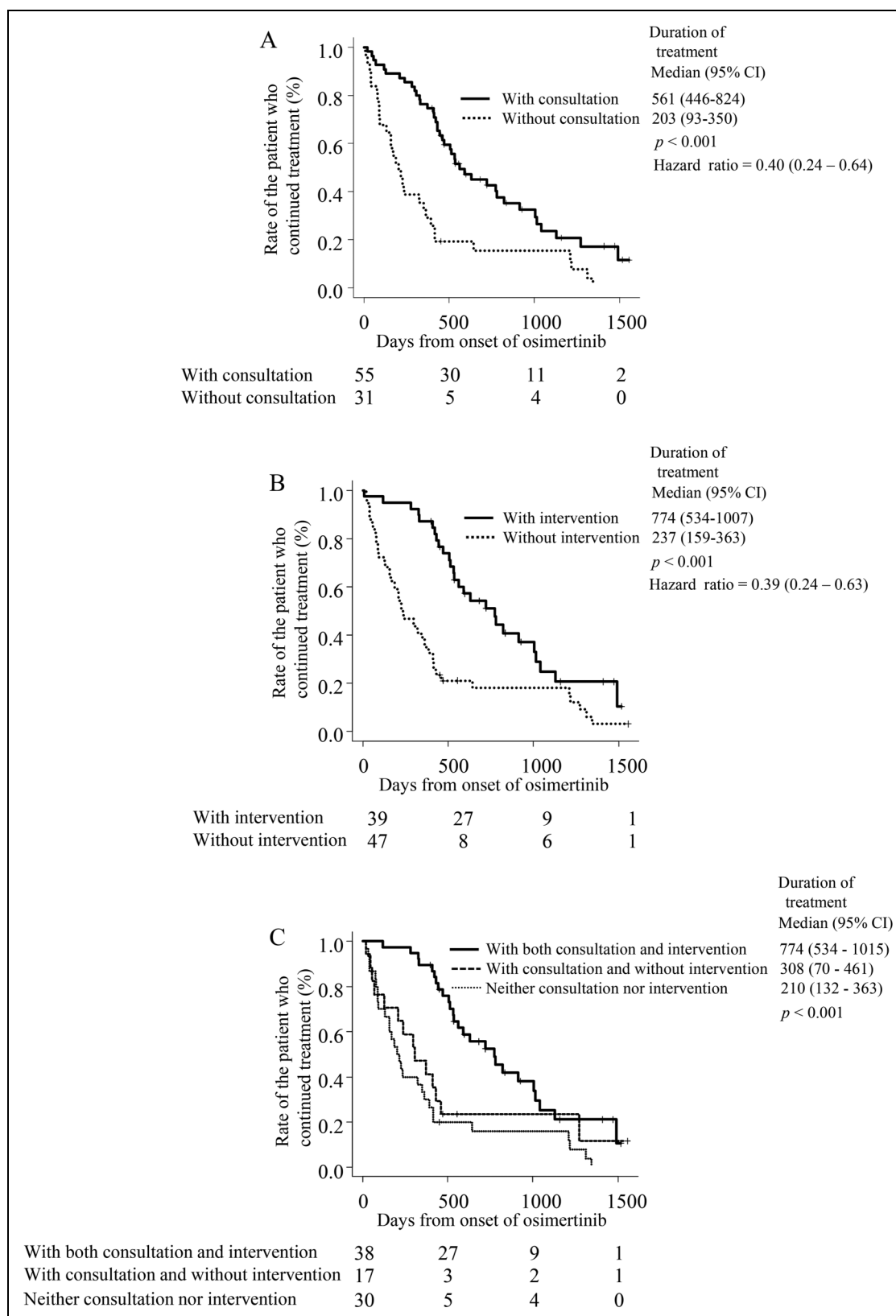


Figure 1. Effects of consultation and pharmacist intervention in the pharmacist-led outpatient clinic on the duration of treatment with osimertinib; consultation (A), intervention (B), and both consultation and intervention (C).

Other reasons for discontinuation were participation in a clinical trial, change to curative radiotherapy, treatment of other concomitant cancers, hypoadrenalism, heart failure and admission to another hospital for treatment of fractures.

Adverse events

Adverse events that were assessed were skin rash, diarrhea, dry skin, paronychia, and oral mucositis, since these adverse events had higher incidence in the FLAURA study. The severity of adverse events was recorded in approximately 90% of patients who consulted a PLOC pharmacist, compared to just 40% in those who did not ($p < 0.001$, Figure 2). Therefore, this study focused on patients who consulted a PLOC pharmacist, and the proportion of patients who experienced each adverse event was compared between with and without pharmacist intervention. The results showed that dry skin and paronychia were significantly less in patients who received pharmacist intervention than in those who did not ($p < 0.001$ and $p = 0.004$, Figure 3).

Details of pharmacist interventions

A total of 93 pharmacist interventions were provided to 39 patients; 88 were for osimertinib-induced adverse events, of which 43 were prescribing suggestions to doctors for adverse events and 45 were instructions to patients for management of adverse events (Table 2). Fourteen prescribing suggestions to doctors and seven instructions to patients were provided for paronychia, and six prescribing suggestions to physicians and fourteen instructions to patients were provided for dry skin. Interventions for paronychia and dry skin, which were higher-grade adverse events, accounted for about 40% of total pharmacist interventions. Five interventions were for other than osimertinib-induced adverse events; including suggestions for laxatives, dose adjustment for osimertinib-rifampicin interaction, and simple suspension method and swallowing jelly for dysphagia.

Discussion

In this study, the effects of PLOC pharmacist consultation and pharmacist interventions on the treatment adherence and management of adverse events with osimertinib were investigated. The results showed that osimertinib treatment duration was significantly longer in patients who consulted a PLOC pharmacist and even longer in those who received pharmacist interventions. The longer duration of osimertinib treatment despite the severity of adverse events in the patients who received pharmacist interventions suggests that the prompt control of adverse events based on the appropriate assessment contributed to continuing osimertinib treatment. This is the first report to show the benefit

of pharmacist interventions in a PLOC for patients with NSCLC who received osimertinib treatment, and it is expected to contribute to further development of osimertinib treatment and healthcare collaboration.

First, for the primary outcome, the duration of osimertinib treatment was prolonged in the patients who consulted a PLOC pharmacist and further prolonged by pharmacist interventions than in those who did not. Regarding the effects of pharmacist interventions, a trend of longer treatment periods for lenvatinib therapy in patients with hepatocellular carcinoma¹¹ and longer progression-free survival for erlotinib therapy in patients with NSCLC¹³ were reported. However, these studies were limited by small sample sizes. Other reports have demonstrated longer treatment durations with sorafenib in hepatocellular carcinoma,⁷ improved completion rates for tegafur, gimeracil, and oteracil in gastric cancer,⁹ and reduced severe adverse events with kinase inhibitors.⁸ The present study suggested that pharmacist interventions also contribute to longer treatment duration in osimertinib therapy for patients with NSCLC.

Patients in the present study who had not previously been treated with a first- or second-generation EGFR-TKI were predominantly included in the patients who consulted a PLOC pharmacist. It cannot be excluded that prior treatment with an EGFR-TKI might have affected the duration of osimertinib therapy, since the median progression-free survival of the second-line osimertinib therapy in the AURA 3 trial was 10.1 months and that of the first-line osimertinib therapy in the FLAURA trial was 18.9 months.^{3,14} Therefore, the duration of osimertinib therapy in patients who had not been previously treated with an EGFR-TKI was analyzed. The results showed that the duration of osimertinib therapy was significantly longer in the patients who consulted a PLOC pharmacist than in those who did not (536 vs 159 days, $p = 0.003$), and in the patients who received pharmacist interventions than in those who did not (590 vs 219 days, $p = 0.001$). These results suggested that PLOC pharmacist consultation and pharmacist interventions contributed to longer duration of osimertinib therapy, irrespective of previous treatment with first- or second-generation EGFR-TKIs.

In the present study, 41% of patients required dose reductions, and 20% of patients discontinued osimertinib treatment due to adverse events. In the final analysis of the FLAURA study, 4% of patients required dose reductions, and 15% of patients discontinued osimertinib treatment due to adverse events.⁴ Although a simple comparison cannot be made, many patients required dose reduction in the present study. A possible reason for this is that osimertinib had been administered even to patients with poor PS in actual clinical practice, even though patients with PS2 or higher were excluded from the FLAURA study. In addition, some patients had osimertinib withdrawn or reduced in dose even for grade 1 or 2 adverse events in actual clinical practice, whereas the withdrawal

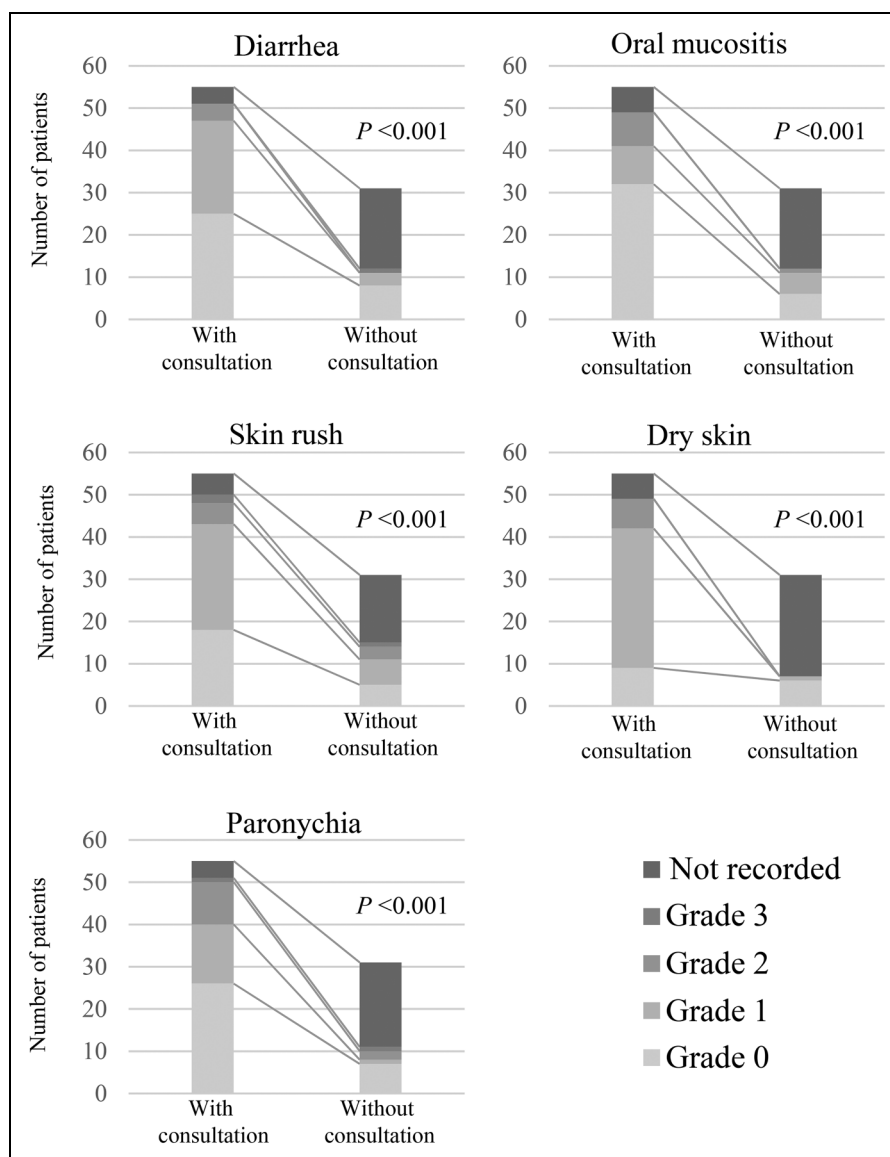


Figure 2. Effect of consultation in the pharmacist-led outpatient clinic on the assessment severity of adverse events.

criteria in the FLAURA study were based on the occurrence of grade 3 or higher adverse events. There were no differences in the rate and reason for osimertinib withdrawal or dose reduction between the patients who consulted a PLOC pharmacist and those who did not. The discontinuation rate during the observation period was significantly lower in patients who consulted a PLOC pharmacist than in those who did not (73% vs 97%, $p=0.008$), but the reasons for discontinuation were not different between the groups. This suggests that PLOC consultation reduces the discontinuation rate without affecting the reasons for withdrawal, dose reduction, and discontinuation due to adverse events.

In the present study, the severity of adverse events was documented in the medical record for only 40% in the

patients who did not consult a PLOC pharmacist. Ono et al. reported that suspected adverse events were more frequently not documented in the medical record in the non-pharmacist intervention group,¹¹ which corresponds with the results of the present study. Since it was difficult to compare the severity of adverse events between the patients who consulted a PLOC pharmacist and those who did not, this study focused on patients who consulted a PLOC pharmacist and compared the severity of adverse events between the patients who received pharmacist intervention and those who did not. The results showed that the severity of paronychia and dry skin was significantly higher in patients who received pharmacist intervention than in those who did not. Grade 1–2 dry skin occurred in 86% of patients, and none developed grade 3 in patients who

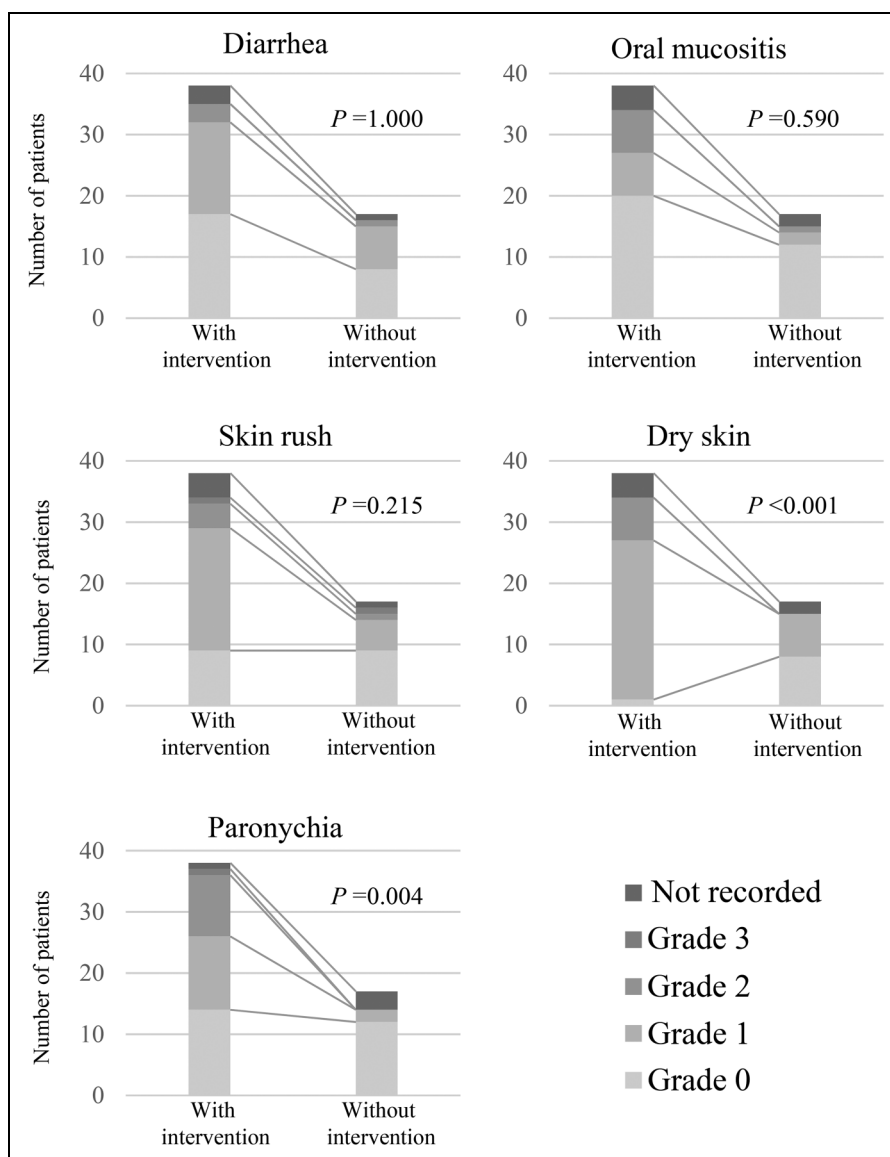


Figure 3. Effect of pharmacist intervention on severity of adverse events.

received pharmacist intervention. Paronychia caused withdrawal and dose reduction in three patients and discontinuation in one patient, whereas dry skin did not cause withdrawal, dose reduction, or discontinuation of osimertinib therapy. Interventions for dry skin and paronychia accounted for about 40% of all pharmacist interventions, which were provided for higher-grade adverse events.

The study has several limitations. First, the severity of adverse events was assessed at the worst grade during the treatment period, and changes in severity before and after the intervention were not evaluated. Interventions were provided to patients who consulted a pharmacist in the PLOC and were assessed by the pharmacist as in need of intervention, and not to patients who did not. Interventions were not randomly assigned, since whether a patient consulted a

PLOC pharmacist was based on the patient's preference or the doctor's request. Second, the duration of treatment based on severity of adverse events was not analyzed, because their severity was not assessed in quite a number of patients who did not visit the PLOC. Third, more patients who received pharmacist's interventions had severe adverse events. The development of skin disorders has been reported to correlate with the efficacy of EGFR-TKI,¹⁵ which might have affected the duration of treatment. Therefore, we conducted subset analysis focused on the patients who developed skin disorders and compared the durations of treatment between patients who consulted a PLOC pharmacist and those who did not. The results showed that a longer duration of treatment in patients who consulted a PLOC pharmacist than in those who did

Table 2. Details of pharmacist interventions.

Interventions	Suggestions to doctors (n)	Instructions to patients (n)
Adverse events		
Paronychia	14	7
Dry skin	6	14
Skin rash	9	10
Oral mucositis	5	4
Diarrhea	1	1
Others (itching, nail deformity, cracks in limbs)	8	9
Other symptoms		
Laxatives for constipation	1	
Menstrual irregularities		1
Dry mouth due to etizolam (change to on demand use)	1	
Osimertinib dose up when rifampicin added	1	
Simple suspension method and swallowing jelly for dysphagia		1

not (590 vs 382 days, $p=0.091$), and a significant prolongation was observed in patients who received pharmacist interventions than those who did not (774 vs 350 days, $p=0.004$). These results suggest that continuous monitoring and appropriate assessment in the PLOC may lead to prompt pharmacist intervention for adverse events and contribute to prolonging the duration of treatment by reducing discontinuations in osimertinib therapy.

Conclusion

PLOC pharmacist consultations and interventions may improve adverse event management and extend osimertinib treatment duration. However, since this was a single-center study with a limited sample size, larger multi-center studies are needed for validation.

Author contribution

ST and IT had full access to all data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. ST and IT designed the study. ST and IT performed the analysis and interpreted the results. ST and IT drafted the manuscript. IT, ST, AK and SI made critical revision of the manuscript for important intellectual content. SI, AK, HT, MO and IT provided administrative, technical, or material supports. All authors critically reviewed and revised the manuscript draft and approved the final version of the manuscript.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical approval and informed consent statements

The ethical review board of the Aichi Medical University School of Medicine approved the study (approval number: 2019-203). Opt-out consent was obtained from participants in this study with the permission of the board because no intervention beyond routine clinical practice was included.

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